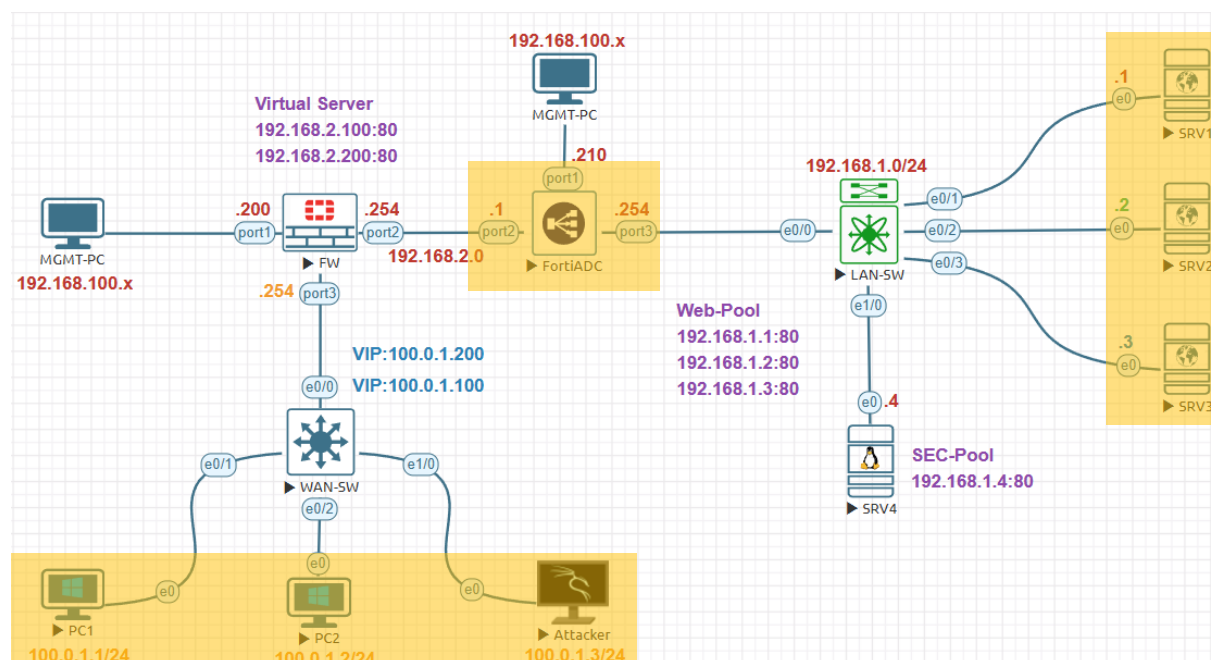


DoS Protection Lab:



Management Subnet	192.168.100.0/24
FortiGate Management IP	192.168.100.200
FortiADC Management IP	192.168.100.210
Internal Servers Subnet	192.168.1.0/24
FortiADC External Subnet	192.168.2.0/24
FortiGate Firewall External	100.0.1.0/24
FortiADC Internal IP Address	192.168.1.254
FortiADC External IP Address	192.168.2.1
FortiGate Firewall Internal IP	192.168.2.254
FortiGate Firewall External IP	100.0.1.254
SRV1 IP Address	192.168.1.1
SRV2 IP Address	192.168.1.2
SRV3 IP Address	192.168.1.3
SRV3 IP Address	192.168.1.4
Virtual Sever-1	192.168.2.100
Virtual Sever-2	192.168.2.200
FortiGate VIP-1	100.0.1.100
FortiGate VIP-2	100.0.1.200
External PC1 IP Address	100.0.1.1
External PC2 IP Address	100.0.1.2
External PC3 IP Address	100.0.1.3

DoS Protection Networking Setting:

Before create DoS Protection Profile let's configure Networking configuration, Click **DoS Protection > Networking > IP Fragmentation Protection**. Change the value if you want

Dashboard	>	IP Fragmentation Protection	TCP SYN Flood Protection	TCP Slow
Security Fabric	>			
FortiView	>	Max Memory Size Limit	4096	
System	>		Default: 4096 Range: 0-4096 (KB)	
Shared Resources	>	Min Memory Size Limit	3072	
Network	>		Default: 3072 Range: 0-4096 (KB)	
Server Load Balance	>	Timeout	30	
Link Load Balance	>		Default: 30 Range: 1-60	
Global Load Balance	>			
Web Application Firewall	>			
Network Security	>			
DoS Protection	▼			
DoS Protection Profile				
Application				
Networking				

Click **DoS Protection > Networking > TCP SYN Protection**. Enable **SYN Cookie** and click **Save**.

IP Fragmentation Protection	TCP SYN Flood Protection	TCP Slow Data Flood Protection	TCP Co
SYN Cookie	☒		
Maximum Half-Open Sockets	100		
	Range: 1-80000 (1=10 connections)		
 Save Refresh			

Next Click **DoS Protection > Networking > TCP Slow Data Flood Protection**. Click the **Create New** to create new TCP Slow Data Flood Protection using following settings.

Name: TCP-SlowData

Status: Enable

Action: Deny

Log: Enable

Severity: High

And finally click **Save**.

TCP Slow Data Flood Protection	
Name	TCP-SlowData
Status	<input checked="" type="button" value="Enable"/> Disable
Probe Interval	30 Default: 30 Range: 1-256
Probe Count	6 Default: 6 Range: 1-256
Action	Pass <input checked="" type="button" value="Deny"/> Period Block
Log	<input checked="" type="button" value="Disable"/> Enable
Severity	Low Medium <input checked="" type="button" value="High"/>
 Save Cancel	

Next Click **DoS Protection > Networking > TCP Connection Access Flood Protection**. Click the **Create New** to create new TCP Connection Access Flood Protection using following settings.

TCP Connection Access Flood Protection	
Name	TCP-Connection
Status	Disable <input checked="" type="button" value="Enable"/>
Limit	1 Default: 0 Range: 0-65535. Limits the amount of TCP requests from a certain IP. 0 means no limit for TCP request.
Action	Pass <input checked="" type="button" value="Deny"/> Period Block
Log	<input checked="" type="button" value="Disable"/> Enable
Severity	Low Medium <input checked="" type="button" value="High"/>
 Save Cancel	

DoS Protection Application Setting:

Next Click **DoS Protection > Application > HTTP Access Limit**. Click the **Create New** to create new HTTP Access Limit using following settings.

The screenshot shows the FortiGate web interface. On the left sidebar, 'DoS Protection' is selected. The main content area shows the 'HTTP Access Limit' configuration page. At the top, there are tabs for 'HTTP Access Limit', 'HTTP Connection Flood', and 'HTTP Request Flood'. Below the tabs are buttons for 'Delete', '+ Create New', and '+ Add Filter'. A table lists the existing HTTP Access Limit entries:

Name	Status	HTTP Request Limit
HTTAccessLimit	Enable	1

Below the table, it says 'Showing 1 to 1 of 1 entries 0 rows selected Show 25 entries'. At the bottom, there is a 'DoS Protection Profile' section with a sub-tab 'Application' highlighted.

Give Name: HTTPAccessLimit, make the Status: Enable HTTP Request Limit: 1 Action: Deny, log: Enable and Severity: High finally click Save.

The screenshot shows the 'HTTP Access Limit' configuration form. The fields are as follows:

- Name: HTTAccessLimit
- Status: Disable **Enable**
- HTTP Request Limit: 1
- Action: Deny
- Log: Disable **Enable**
- Severity: Low Medium **High**

At the bottom right, there is a red arrow pointing to the 'Save' button, and a 'Cancel' button next to it.

Next Click **DoS Protection > Application > HTTP Connection Flood**. Click the **Create New** to create new HTTP Connection Flood using following settings.

HTTP Connection Flood	
Name	HTTPConnectionFlood
Status	Disable <input checked="" type="button" value="Enable"/>
TCP Connection Number Limit	1 Default: 1 Range: 1-1024. Limits the number of TCP connections with the same session cookie.
Action	Deny
Block Number	100000 Default: 100000 Range: 1-1000000
Log	Disable <input checked="" type="button" value="Enable"/>
Severity	Low Medium <input checked="" type="button" value="High"/>



Next Click **DoS Protection > Application > HTTP Request Flood**. Click the **Create New** to create new HTTP Request Flood using following settings.

HTTP Request Flood	
Name	HTTPRequestFlood
Status	Disable <input checked="" type="button" value="Enable"/>
HTTP Request Limit	1 Default: 0 Range: 0-65535. Limits the number of HTTP requests per second with the same session cookie. 0 means no limit for HTTP request.
Action	Deny
Block Number	200000 Default: 200000 Range: 1-1000000
Log	<input checked="" type="button" value="Disable"/> Enable
Severity	Low Medium <input checked="" type="button" value="High"/>



DoS Protection Profile:

Now let's create DoS Profile, Click **DoS Protection > DoS Protection Profile**. Click the **Create New** to create new DoS Protection Profile using following settings.

The screenshot shows the FortiGate web interface. On the left is a sidebar menu with various system components. The 'DoS Protection' option is highlighted with a red box. Below it, the 'DoS Protection Profile' sub-menu is also highlighted with a red box. On the right, the main content area shows a table with columns for Name, HTTP Connection Flood, and HTTP Access Limit. Above the table, there are buttons for 'Delete', '+ Create New' (highlighted with a red box), and '+ Add Filter'. Below the table, it says 'Showing 1 to 1 of 1 entries 0 rows selected Show 25 entries'.

Name	HTTP Connection Flood	HTTP Access Limit

Showing 1 to 1 of 1 entries 0 rows selected Show 25 entries

Give it name in this case **DOS-Profile** from dropdown choose all the profile previously create and click **Save**.

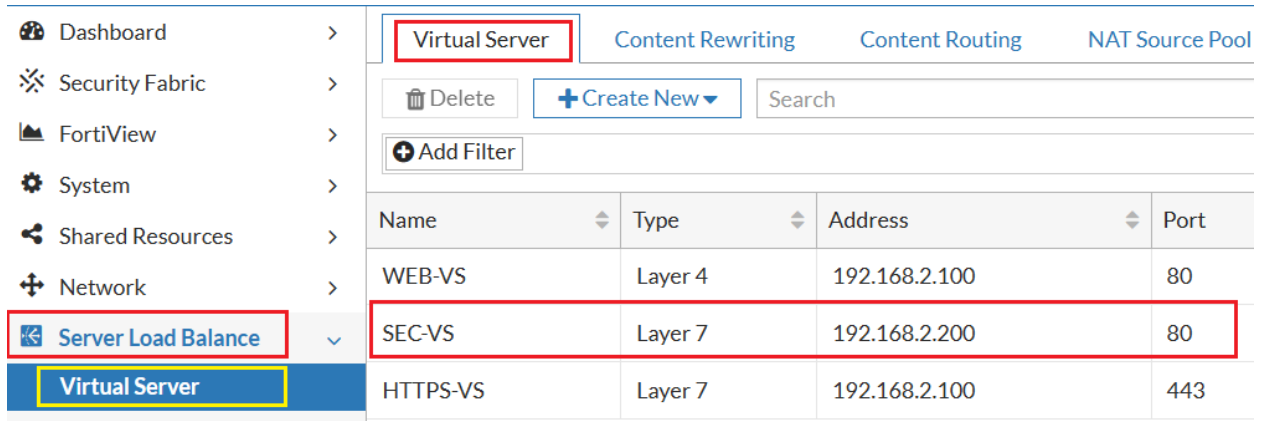
The screenshot shows the 'DoS Protection Profile' configuration form. The form has several fields, each with a dropdown menu. The 'Name' field is set to 'DOS-Profile'. The other fields are 'HTTP Connection Flood', 'HTTP Access Limit', 'HTTP Request Flood', 'TCP Slow Data Flood Protection', and 'TCP Connection Access Flood Protection'. Each of these fields has a dropdown menu with a red box around it, indicating the selection process. At the bottom right, there are 'Save' and 'Cancel' buttons, with a red arrow pointing to the 'Save' button.

DoS Protection Profile	
Name	DOS-Profile
HTTP Connection Flood	HTTPConnectionFlood
HTTP Access Limit	HTTPAccessLimit
HTTP Request Flood	HTTPRequestFlood
TCP Slow Data Flood Protection	TCP-SlowData
TCP Connection Access Flood Protection	TCP-Connection

Save **Cancel**

Attach DoS Profile to VS:

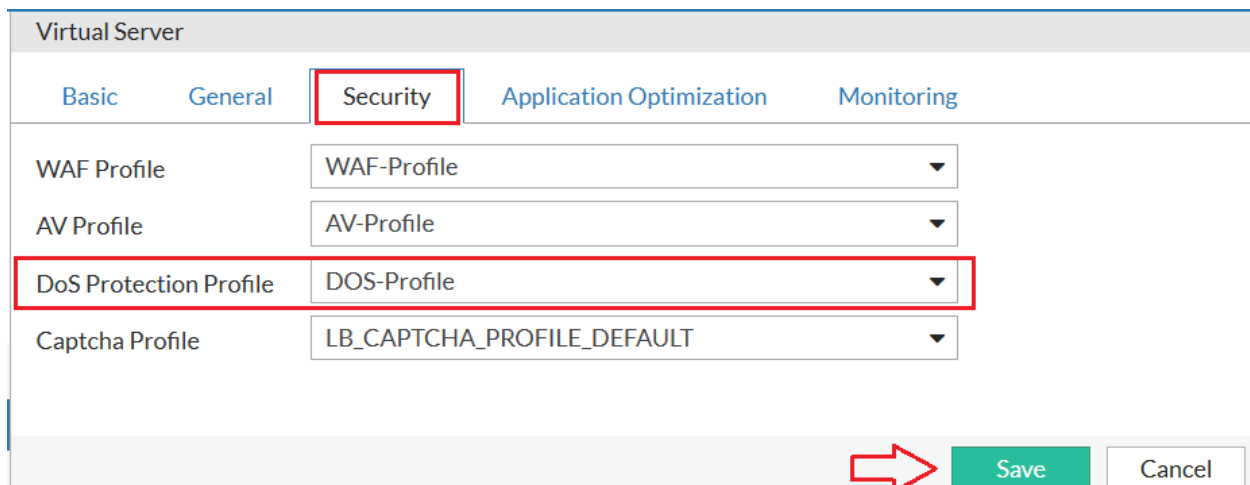
Click **Server Load Balance > Virtual Server > Virtual Server**. Click the edit icon or double click to edit the new virtual server **SEC-VS**.



The screenshot shows the FortiGate web interface. On the left sidebar, 'Server Load Balance' and 'Virtual Server' are highlighted. The main content area shows the 'Virtual Server' configuration page. At the top, there are tabs for 'Virtual Server', 'Content Rewriting', 'Content Routing', and 'NAT Source Pool'. Below these tabs are buttons for 'Delete', '+ Create New', and a search bar. There is also an 'Add Filter' button. A table lists the virtual servers:

Name	Type	Address	Port
WEB-VS	Layer 4	192.168.2.100	80
SEC-VS	Layer 7	192.168.2.200	80
HTTPS-VS	Layer 7	192.168.2.100	443

On the **Security** tab, change the **DoS Protection Profile** setting to **DOS-Profile**, and then click **Save**.



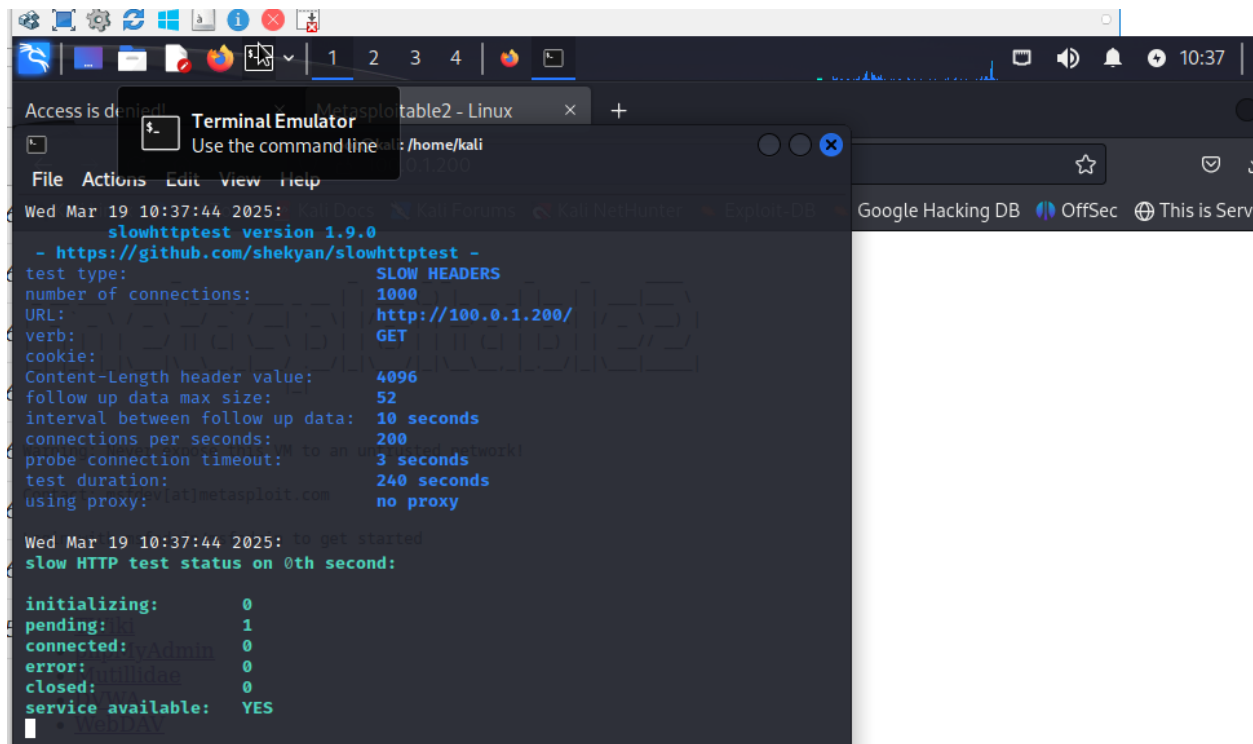
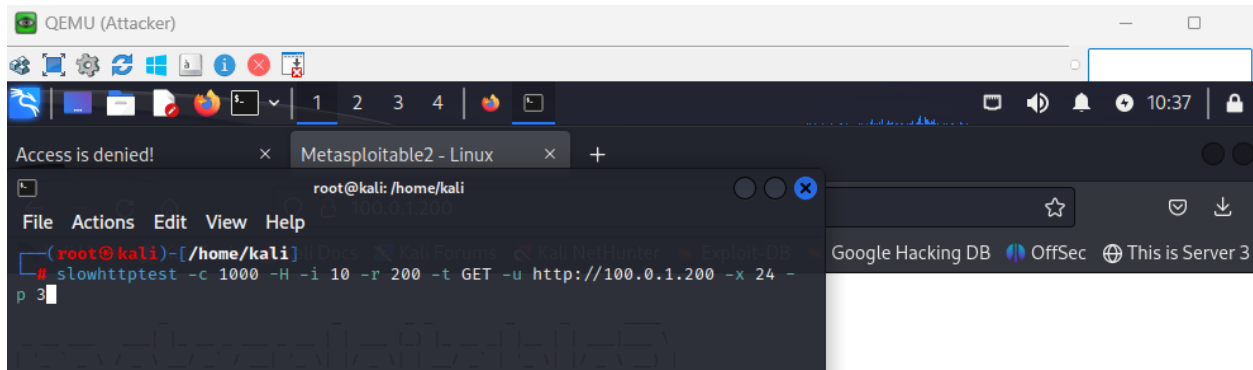
The screenshot shows the 'Virtual Server' configuration page with the 'Security' tab selected. The 'DoS Protection Profile' is set to 'DOS-Profile'. The 'Save' button is highlighted with a red arrow.

Virtual Server	
WAF Profile	WAF-Profile
AV Profile	AV-Profile
DoS Protection Profile	DOS-Profile
Captcha Profile	LB_CAPTCHA_PROFILE_DEFAULT

Testing & Verification:

On the External Kali Linux, open Terminal and run below command:

slowhttptest -c 1000 -H -i 10 -r 200 -t GET -u <http://100.0.1.200> -x 24 -p 3



Navigate to **Log & Report>Security Log >** choose from dropdown **DDoS** to verify the logs

Download

Refresh

DDoS

25/03/07 21:19:19 - 25/03/19 15:43:20

Add Filter

Date	Time	Count	Source	Destination
2025-03-19	17:37:48	1001	0.0.0.0	192.168.2.200
2025-03-19	17:36:53	2004	0.0.0.0	192.168.2.200
2025-03-19	17:36:40	132	 100.0.1.3	192.168.2.200
2025-03-19	17:36:37	1005	0.0.0.0	192.168.2.200
2025-03-19	17:36:09	1	 100.0.1.3	192.168.2.200
2025-03-19	14:56:22	1971	0.0.0.0	192.168.2.200
2025-03-19	14:56:04	1000	0.0.0.0	192.168.2.200
2025-03-19	14:56:01	2025	0.0.0.0	192.168.2.200
2025-03-19	14:55:38	1020	0.0.0.0	192.168.2.200